

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Experiments

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)</i>		
Student Name:	Sk. Fawaz Ahamad	Reg. No.:	192425390
Section / Batch:	2024	Date:	11-08-2026

Code of Conduct

I Sk. Fawaz Ahamad / 192425390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk. Fawaz Ahamad

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts	
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation	
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools	
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results	
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation	

1. Preprocessing Impact Analysis

Design and perform an experiment to study the impact of preprocessing techniques such as noise removal and normalization on image quality and feature detection. Explain the theoretical concepts behind preprocessing methods and justify the selection of techniques used. Implement the preprocessing process using suitable software/tools, document the workflow and outputs systematically, and analyze how preprocessing influences feature detection accuracy and reliability.

2. Image Representation Study

Conduct an experiment to convert a given image into grayscale and binary representations. Explain the significance of different image representations in computer vision applications. Design the conversion procedure using appropriate tools, document the generated outputs clearly, and analyze how image representation affects edge detection and feature extraction performance.

3. Edge Detection Comparison

Design an experiment to compare Sobel and Canny edge detection techniques on the same input image. Explain the underlying concepts and operational principles of both methods. Apply the techniques using suitable tools/software, document the detected edges systematically, and analyze the comparative effectiveness of each method for different image conditions and applications.

4. Harris Corner Detection Analysis

Perform Harris corner detection on images containing varying textures and patterns. Explain the concept of corner detection and its importance in computer vision tasks. Use appropriate parameters and tools to execute the experiment, document the detected corner points clearly, and analyze the distribution, accuracy, and significance of the detected corners.

5. Effect of Noise on Corner Detection

Design an experiment to study the effect of noise on Harris corner detection performance. Explain the purpose of introducing noise and its influence on image features. Add noise systematically, perform corner detection using suitable tools, document observations for original and noisy images, and analyze the impact of noise along with possible improvement techniques.

6. Hough Transform for Shape Detection

Conduct an experiment using the Hough Transform to detect lines or circles in an image. Explain the principles and significance of the Hough Transform in shape detection. Execute the detection process using appropriate tools, record the detected shapes and their parameters, and analyze the detection accuracy under different image conditions.

7. Feature Matching using SIFT

Perform feature extraction and matching between two images using the SIFT algorithm. Explain the concept of feature matching and scale-invariant features. Use suitable

software/tools to extract and match features, document the matching key points systematically, and analyze the robustness of SIFT under variations in scale, rotation, and illumination.

8. PCA for Feature Reduction

Design and implement an experiment using Principal Component Analysis (PCA) for dimensionality reduction of extracted image features. Explain the importance of feature reduction in machine learning and computer vision applications. Apply PCA using appropriate tools, document the feature dimensions before and after reduction, and analyze its impact on computational efficiency and feature representation quality.

9. Preprocessing vs Feature Extraction

Conduct a comparative study of feature extraction performed with and without preprocessing. Explain the relationship between preprocessing and feature extraction performance. Execute both approaches using suitable tools, document the observations systematically, and analyze the differences in feature quality, detection accuracy, and overall system performance.

10. Integrated Feature Detection Pipeline

Design and implement a complete image processing pipeline consisting of preprocessing, edge detection, and feature extraction stages. Explain the purpose and workflow of the integrated pipeline. Execute each stage systematically using appropriate tools, document intermediate and final outputs clearly, and analyze the contribution and importance of each stage in achieving accurate feature detection results.

11. Effect of Smoothing on Edge Detection

Design and perform an experiment to study the impact of smoothing techniques on edge detection performance. Explain the theoretical purpose of smoothing in reducing noise and improving edge continuity. Apply smoothing and edge detection methods using appropriate tools, document outputs before and after smoothing systematically, and analyze how smoothing affects edge clarity, continuity, and noise suppression.

12. Gradient-Based Edge Detection Study

Conduct an experiment using gradient-based edge detection methods to identify image boundaries. Explain the principles of gradient operators and their role in edge detection. Implement the method using suitable tools/software, record gradient values and detected edges systematically, and analyze edge strength, accuracy, and sensitivity to image variations.

13. Prewitt vs Sobel Comparison

Design a comparative study of Prewitt and Sobel edge detection operators. Explain the conceptual differences and applications of both methods. Execute both operators accurately on the same image using appropriate tools, document the outputs clearly, and analyze their performance in detecting edges under varying image conditions.

14. Effect of Image Scaling on Feature Detection

Perform an experiment to analyze how image scaling affects feature detection accuracy. Explain the importance of scale in computer vision and feature extraction tasks. Resize images to

different scales, apply feature detection techniques, document detected features systematically, and analyze the impact of scaling on feature consistency and detection performance.

15. Rotation Impact on Feature Matching

Conduct an experiment to evaluate the effect of image rotation on feature matching accuracy. Explain the significance of rotation invariance in feature matching algorithms. Rotate images at different angles, perform feature matching using suitable tools, document matching results clearly, and analyze the robustness of the method against rotational transformations.

16. Threshold Effect in Edge Detection

Design an experiment to study the influence of threshold values on edge detection quality. Explain the importance of thresholding in distinguishing meaningful edges from noise. Apply edge detection using different threshold values, record the outputs systematically, and analyze how threshold selection affects edge continuity, accuracy, and noise sensitivity.

17. Noise Removal Comparison

Perform a comparative study of different noise removal techniques applied to images. Explain the objectives and principles of commonly used filtering methods. Implement multiple noise reduction filters using suitable tools, document the outputs clearly, and analyze the effectiveness of each technique in preserving image details while removing noise.

18. Feature Detection in Noisy Images

Conduct an experiment to analyze feature detection performance on noisy images. Explain how image noise affects feature extraction and detection accuracy. Apply feature detection techniques on noisy images using appropriate tools, record detected features systematically, and analyze the impact of noise on feature reliability and stability.

19. Edge Detection under Illumination Changes

Design an experiment to study the effect of illumination variation on edge detection performance. Explain the relationship between lighting conditions and edge extraction accuracy. Apply edge detection methods on images with varying illumination levels, document the outputs clearly, and analyze the robustness of the techniques under different lighting conditions.

20. Corner Detection Parameter Study

Perform a parameter analysis study for Harris corner detection. Explain the significance of parameters in influencing corner detection accuracy. Execute the corner detection algorithm using different parameter values, record the detected corners systematically, and analyze how parameter variations affect detection sensitivity and performance.

21. Hough Transform Parameter Analysis

Conduct an experiment to evaluate the impact of parameter variations in the Hough Transform. Explain the role of parameters in shape detection accuracy. Apply the Hough Transform using different parameter settings, document the detected shapes and outputs clearly, and analyze the sensitivity and reliability of the detection process.

22. Feature Matching Accuracy Evaluation

Design an experiment to evaluate the accuracy of feature matching between images. Explain the objectives and significance of feature matching evaluation. Perform feature matching using suitable algorithms and tools, record matching points systematically, and analyze matching accuracy, false matches, and overall reliability.

23. PCA Component Selection Study

Perform a study on the effect of varying the number of principal components in PCA. Explain the importance of component selection in dimensionality reduction. Execute PCA with different numbers of components, document feature dimensions and outputs clearly, and analyze the trade-off between dimensionality reduction and information preservation.

24. Edge Detection vs Corner Detection

Conduct a comparative analysis of edge detection and corner detection techniques on the same image. Explain the objectives and differences between both approaches. Apply suitable methods using appropriate tools, document the detected features systematically, and analyze the types of features captured and their relevance in computer vision applications.

25. Multi-Stage Pipeline Analysis

Design and implement a multi-stage image processing pipeline consisting of preprocessing, edge detection, and feature extraction. Explain the objectives and workflow of the pipeline. Execute each stage systematically using suitable tools, document outputs at every stage clearly, and analyze the contribution and effectiveness of each stage in improving overall system performance.

26. Feature Detection Density Analysis

Perform an experiment to analyze the density of detected image features and its impact on performance. Explain the importance of feature density in computer vision tasks. Execute feature detection using appropriate tools, record the number and distribution of detected features systematically, and analyze the relationship between feature density, computational complexity, and detection accuracy.

27. Effect of Blurring on Feature Detection

Conduct an experiment to study the influence of image blurring on feature detection. Explain the purpose of blurring and its effect on image details. Apply blur filters followed by feature detection techniques, document the outputs clearly, and analyze how blurring affects feature visibility, accuracy, and robustness.

28. PCA Visualization Study

Perform a visualization study of features before and after applying PCA. Explain the significance of dimensionality reduction in feature analysis. Execute PCA using suitable tools, document visual outputs systematically, and analyze how PCA transforms and compresses feature representations while preserving important information.

29. Feature Matching under Noise and Rotation

Design an experiment to evaluate feature matching performance under combined noise and rotation conditions. Explain the challenges introduced by image transformations and noise. Apply noise and rotational transformations, perform feature matching using suitable tools, document the matching results clearly, and analyze the robustness and reliability of the matching technique.

30. End-to-End Vision System Experiment

Design and implement a complete computer vision pipeline involving preprocessing, feature detection, feature matching, and PCA-based dimensionality reduction. Explain the objectives and workflow of the integrated system. Execute all stages systematically using appropriate tools, document outputs at each stage clearly, and analyze the overall system performance, limitations, and bottlenecks.

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

(SIMATS)

Chennai, Tamil Nadu

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. Tech – Artificial Intelligence & Machine Learning

CO3-AT1

Scenario Based Question

Category	Details
Submitted By	Sk. Fawaz Ahamad
Register No.	192425390
Program	B. Tech Artificial Intelligence & Machine Learning
Year & Semester	III Year
Department	Computer Science and Engineering
Submitted To	Faculty Name: S. christy
Subject	Computer Vision
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COMPUTER VISION EXPERIMENT – 24

Experiment Title

Image Segmentation – Under-Segmentation

1. Aim

To study the problem of **under-segmentation** in image segmentation, identify the causes of incorrect merging of image regions, and understand how **Watershed Segmentation** can be used to separate connected or touching objects.

2. Objective

The objectives of this experiment are:

- To understand the concept of under-segmentation.
- To identify incorrectly merged regions in an image.
- To study the major causes of under-segmentation.
- To understand the separation problem between touching objects.
- To select a suitable segmentation technique for separating the regions.
- To analyze the improvement obtained after segmentation.

3. Introduction

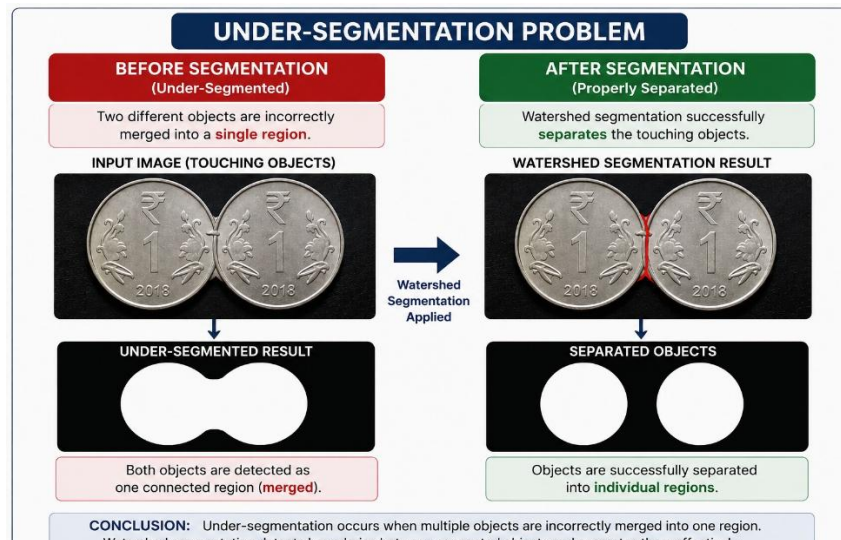
Image segmentation is the process of dividing an image into meaningful regions or objects. It is an important step in computer vision because it helps a system identify and analyze individual objects present in an image.

Under-segmentation occurs when two or more different objects or regions are incorrectly combined into a single segment. As a result, the segmentation does not correctly represent the individual objects in the image.

For example, when two coins are touching each other, a segmentation method may detect both coins as one large region instead of identifying them separately.

4. Under-Segmentation Problem

In under-segmentation, the number of detected regions is **less than the actual number of meaningful objects**.



5. Causes of Under-Segmentation: -

The main causes are:

1. Similar Intensity or Colour

Neighbouring objects may have similar brightness or colour, making it difficult to distinguish them.

2. Weak Boundaries

If the boundary between two objects is not clearly visible, the segmentation algorithm may merge them.

3. Touching Objects

Two objects touching each other may be detected as one connected component.

4. Image Noise

Noise can distort object boundaries and affect segmentation accuracy.

5. Incorrect Threshold

An unsuitable threshold value may cause neighbouring regions to become part of the same segment.

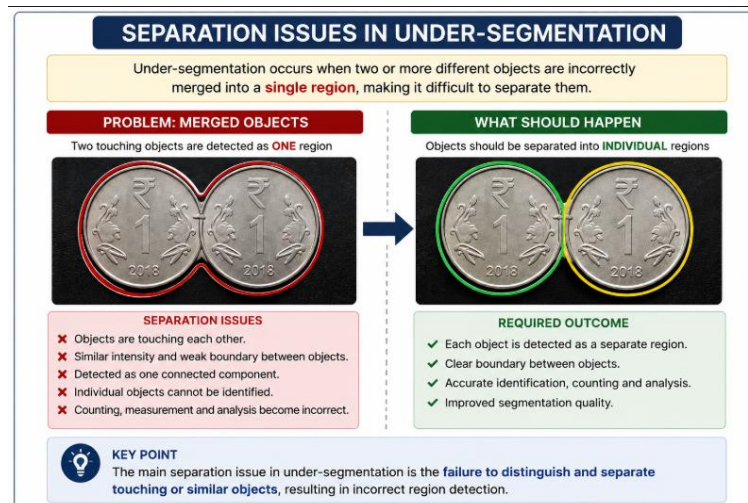
6. Poor Image Quality

Blurred or low-resolution images may not contain enough information to clearly separate objects.

6. Separation Issues

The major separation issue in under-segmentation is that **individual objects cannot be distinguished from each other**.

For example, consider two touching coins:



This can cause problems in applications such as:

- Object detection
- Medical image analysis
- Industrial inspection
- Traffic monitoring
- Counting objects
- Character recognition

7. Suitable Solution – Watershed Segmentation

A suitable technique for solving the separation problem is **Watershed Segmentation**.

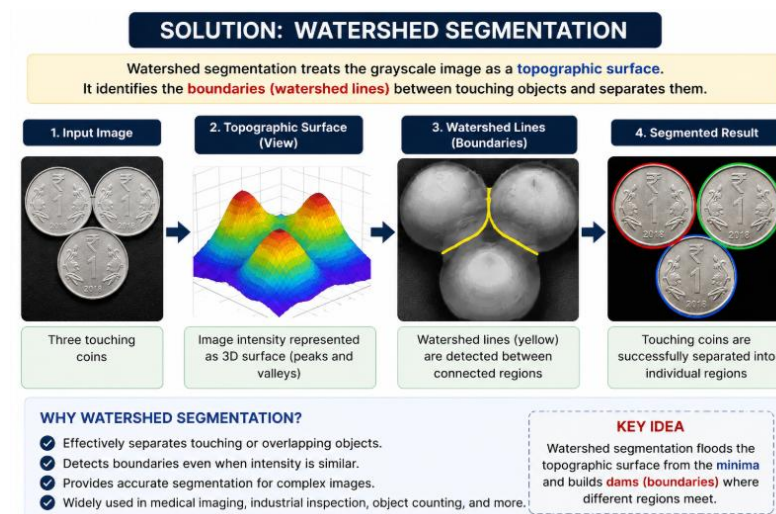
Watershed segmentation is a region-based image segmentation technique. It considers an image as a **topographic surface**, where bright and dark regions represent different heights.

The technique identifies boundaries between connected regions and can therefore separate objects that are touching or connected.

Why Watershed?

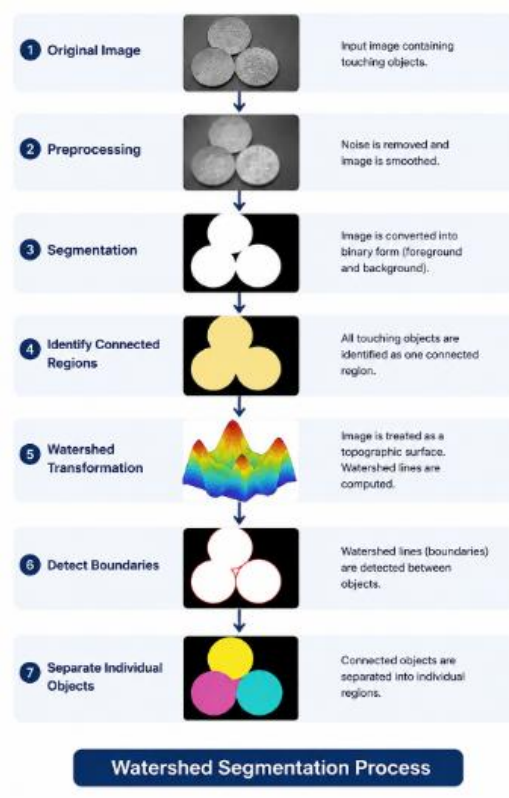
Watershed segmentation is particularly useful when:

- Objects are touching.
- Objects have similar intensity.
- Connected components need to be separated.
- Clear boundaries between objects are required.



8. Working Principle

The general process can be represented as:



9. Experimental Observation

Consider an image containing two or more touching objects.

Before Segmentation

The objects are connected and may appear as **one single region**.

Observation:

The segmentation method fails to identify the individual objects separately.

After Applying Watershed Concept

A boundary is introduced between the connected objects.

Observation:

The previously merged regions are separated into individual objects.

10. Result

The under-segmentation problem was analyzed successfully.

The experiment shows that when multiple objects are incorrectly merged into one region, Watershed Segmentation can identify boundaries between connected objects and separate them into individual regions.

11. Result Comparison

Parameter	Before Separation	After Watershed
Number of regions	Fewer than actual	Closer to actual
Object separation	Poor	Improved
Connected objects	Merged	Separated
Boundaries	Unclear	Clearly identified
Segmentation quality	Low	Improved

12. Advantages of Watershed Segmentation

- Separates touching objects effectively.
- Detects boundaries between connected regions.
- Useful for object counting.
- Suitable for medical and industrial images.
- Helps improve segmentation accuracy.

Limitation

Watershed segmentation can sometimes produce **over-segmentation** if the image contains too many unwanted local regions or noise. Therefore, suitable preprocessing is often required.

13. Applications

Under-segmentation analysis and object separation are useful in:

Medical Imaging

Separating touching cells, tissues, or organs.

Industrial Inspection

Identifying individual components or defects.

Object Counting

Counting objects that are touching each other.

Traffic Analysis

Separating vehicles that appear close together.

Computer Vision

Improving object detection and recognition.

14. Conclusion

Under-segmentation occurs when two or more different image regions are incorrectly merged into a single segment. The problem is mainly caused by similar intensity or colour, weak boundaries, touching objects, noise, and unsuitable segmentation parameters.

Watershed Segmentation provides an effective approach for separating connected objects by identifying boundaries between them. Therefore, it is a suitable technique for reducing under-segmentation in images containing touching or overlapping regions.